

Reasoning about Welfare-Affecting Capabilities in Concurrent Games

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Strategic ability is not only about achieving goals

In multi-agent systems, a coalition's choices may also affect the welfare of other agents.

Classical ATL/CL asks

- Can coalition C ensure φ ?
- Can C keep φ true forever?

We also ask

- Can C benefit coalition B ?
- Can C harm coalition H ?
- Can it do so while ensuring φ ?

Who is a potential benefactor, and who is a danger?

Two Conceptual Ingredients

Harm and benefit

We follow a non-comparative intuition:

- harm: bringing about a bad outcome;
- benefit: bringing about a good outcome.

Good and bad outcomes are represented through agents' preferences.

Danger

We follow a modal, non-probabilistic intuition:

- a coalition is dangerous if it has the potential to harm;
- potential means existence of a suitable strategy.

strategic capability + preference over outcomes $\Rightarrow$ welfare-affecting capability

Main Contributions

- ➊ A semantics based on concurrent games enriched with agents' preferences over computations.
- ➋ New ATL and CL modalities for welfare-affecting strategic capabilities:

$$\langle\langle C:B,H \rangle\rangle\gamma$$

“ C has a strategy that benefits B , harms H , and guarantees γ .”

- ➌ Formal definitions of:

$$\text{Benef}(C, B), \quad \text{Danger}(C, H).$$

- ➍ Technical results:
 - sound and complete axiomatization for the until-free fragment;
 - polynomial embedding into standard ATL/CL with special atoms marking benefited and harmed agents;
 - tight satisfiability complexity: EXPTIME-complete for WA-ATL, PSPACE-complete for WA-CL.

Models: Concurrent Games with Preferences

A model is a concurrent game model with regular preferences:

$$M = (P, \mathcal{V})$$

where P contains:

- states and action profiles;
- a transition relation induced by joint actions;
- for each agent i and state s , a preference order

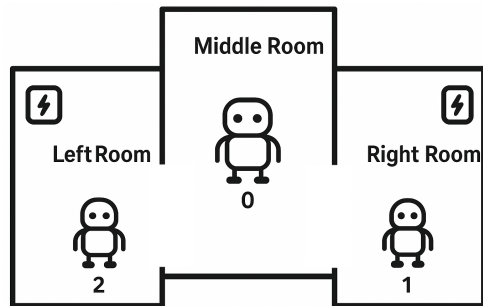
$$\preceq_{i,s}$$

over computations starting at s .

Regular preferences

We assume stability over time, non-flatness, and existence of best and worst computations. These assumptions support the strong notions of benefit and harm.

Running Example: Robots and Charging Rooms



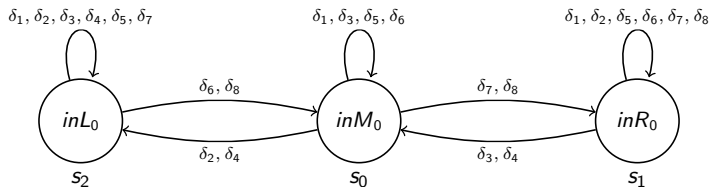
- Robot 0 starts in the middle room.
- It wants to reach a charging station on the left or right.
- Robots 1 and 2 guard the right and left passages.
- Each guard can block or free its passage.

Preference profile

Robot 0 prefers computations where it eventually reaches a charging station.

Robots 1 and 2 prefer computations where robot 0 never reaches one.

The Game Structure



$$\delta_1 = (0:le, 1:bl, 2:bl) \quad \delta_2 = (0:le, 1:bl, 2:fr) \quad \delta_7 = (0:ri, 1:fr, 2:bl) \quad \delta_8 = (0:ri, 1:fr, 2:fr)$$

The relevant strategic tension is: guards can jointly keep robot 0 in the middle.

Strong Benefit and Strong Harm

Let $\mathcal{O}_F(s, \mathcal{F}_C)$ be the computations compatible with strategy $\mathcal{F}_C$ from s .

Strongly beneficial for B

$$\mathcal{F}_C \in \text{StrSB}_M^C(s, B) \quad \text{iff} \quad \mathcal{O}_F(s, \mathcal{F}_C) \subseteq \text{Best}_{i,s} \text{ for every } i \in B.$$

Strongly harmful for H

$$\mathcal{F}_C \in \text{StrSH}_M^C(s, H) \quad \text{iff} \quad \mathcal{O}_F(s, \mathcal{F}_C) \subseteq \text{Worst}_{i,s} \text{ for every } i \in H.$$

Strong benefit/harm means guaranteeing best/worst outcomes for the affected coalition.

Example: Blocking as Harm

Consider the strategy of coalition $\{1, 2\}$:

$$\mathcal{F}_{\{1,2\}}(\lambda) = (1:bl, 2:bl) \quad \text{for every history } \lambda.$$

Then at the initial state s_0 :

$$\mathcal{F}_{\{1,2\}} \in StrSH_{M_{rs}}^{\{1,2\}}(s_0, \{0\}) \quad \text{and} \quad \mathcal{F}_{\{1,2\}} \in StrSB_{M_{rs}}^{\{1,2\}}(s_0, \{1, 2\}).$$

Reading

The same strategy is harmful to robot 0 and beneficial to the guarding coalition.

$$\begin{aligned}\varphi, \psi &::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \langle\langle C:B, H \rangle\rangle\gamma \\ \gamma &::= X\varphi \mid G\varphi \mid \varphi U \psi\end{aligned}$$

Key modality

$$\langle\langle C:B, H \rangle\rangle\gamma$$

Coalition C has a strategy which is beneficial for B , harmful to H , and guarantees temporal property γ .

Standard ATL is recovered as the special case:

$$\langle\langle C \rangle\rangle\gamma \equiv \langle\langle C:\emptyset, \emptyset \rangle\rangle\gamma.$$

$$(M, s) \models \langle\langle C:B, H \rangle\rangle \gamma$$

iff there exists a strategy

$$\mathcal{F}_C \in \text{StrSB}_M^C(s, B) \cap \text{StrSH}_M^C(s, H)$$

such that every generated computation satisfies γ :

$$\forall \lambda \in \mathcal{O}_F(s, \mathcal{F}_C), \quad (M, \lambda) \models \gamma.$$

Three roles in one operator

- C : the acting coalition;
- B : the coalition benefited by the strategy;
- H : the coalition harmed by the strategy.

Potential Benefactor and Danger

The language gives compact definitions of the target notions.

Potential benefactor

$$\text{Benef}(C, B) \equiv \langle\langle C:B, \emptyset \rangle\rangle \text{XT}.$$

Coalition C has a strategy beneficial for B .

Danger

$$\text{Danger}(C, H) \equiv \langle\langle C:\emptyset, H \rangle\rangle \text{XT}.$$

Coalition C has a strategy harmful to H .

In the robot model:

$$(M_{rs}, s_0) \models \text{Benef}(\{1, 2\}, \{1, 2\}) \wedge \text{Danger}(\{1, 2\}, \{0\}).$$

Reasoning About Changing Danger

Nested modalities let us express danger-enabling and danger-disabling choices.

In the robot model:

$$(M_{rs}, s_0) \models \langle\langle\{0, 1\}:\{0\}, \emptyset\rangle\rangle XDanger(\{0\}, \{1, 2\}) \\ \wedge \neg Danger(\{0\}, \{1, 2\}).$$

Intuition

Robot 0 is not initially dangerous to the guards. But coalition $\{0, 1\}$ can choose a beneficial strategy for robot 0 that makes robot 0 dangerous at the next state.

We axiomatize the until-free, or safety, fragment:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \langle\langle C:B,H \rangle\rangle X\varphi \mid \langle\langle C:B,H \rangle\rangle G\varphi.$$

Inference rules

$$\frac{\varphi \quad \varphi \longrightarrow \psi}{\psi} \quad (\text{MP})$$

$$\frac{\varphi \leftrightarrow \psi}{\langle\langle C:B,H \rangle\rangle X\varphi \leftrightarrow \langle\langle C:B,H \rangle\rangle X\psi} \quad (\text{RE})$$

$$\frac{\varphi}{\langle\langle C \rangle\rangle G\varphi} \quad (\text{Gen})$$

Core axiom schemata

All tautologies of propositional logic	(TAUT)
$\neg \langle\langle C:B, H \rangle\rangle X \perp$	(NAAA)
$\langle\langle C:B, H \rangle\rangle X(\varphi \wedge \psi) \longrightarrow \langle\langle C:B, H \rangle\rangle X\varphi$	(MG)
$\langle\langle C:B, H \rangle\rangle X\varphi \longrightarrow \langle\langle C':B', H' \rangle\rangle X\varphi \quad \text{for } C \subseteq C', B \supseteq B', H \supseteq H'$	(MC)
$\langle\langle \emptyset \rangle\rangle X \top$	(NEI)
$(\langle\langle C:B, H \rangle\rangle X\varphi \wedge \langle\langle C':B', H' \rangle\rangle X\psi) \longrightarrow \langle\langle C \cup C':B \cup B', H \cup H' \rangle\rangle X(\varphi \wedge \psi) \quad \text{for } C \cap C' = \emptyset$	(Sup)
$\langle\langle C:B, H \rangle\rangle X(\varphi \vee \psi) \longrightarrow (\langle\langle C:B, H \rangle\rangle X\varphi \vee \langle\langle \text{AGT}:B, H \rangle\rangle X\psi)$	(Cro)
$\langle\langle C:B, H \rangle\rangle G\varphi \leftrightarrow \langle\langle C:B, H \rangle\rangle X(\varphi \wedge \langle\langle C:B, H \rangle\rangle G\varphi)$	(FP _G)
$\langle\langle \emptyset \rangle\rangle G(\psi \longrightarrow \langle\langle C:B, H \rangle\rangle X(\varphi \wedge \psi)) \longrightarrow \langle\langle \emptyset \rangle\rangle G(\psi \longrightarrow \langle\langle C:B, H \rangle\rangle G\varphi)$	(GFP _G)
$\langle\langle \text{AGT}:\emptyset, \{a\} \rangle\rangle X \top$	(WFP)
$\langle\langle \text{AGT}:\{a\}, \emptyset \rangle\rangle X \top$	(CWFP)
$\neg \langle\langle C:B, H \rangle\rangle X \top \quad \text{if } B \cap H \neq \emptyset$	(BHC)

Theorem

For every $\varphi \in \mathcal{L}_{\text{WA-ATL}^{\text{U}}}$:

φ is valid over CGMRPs iff $\vdash \varphi$.

- The proof uses canonical models built under a formula context.
- The context tracks the welfare-affecting modality currently being realized.

Why a Contextual Canonical Model?

Standard ATL-style canonical constructions are not enough because:

$$\langle\langle C:B, H \rangle\rangle \gamma$$

constrains both strategic outcomes and welfare status.

Strategic part

The canonical action must realize the temporal guarantee γ .

Preference part

The same action must generate best computations for B and worst computations for H .

Formula contexts synchronize these two requirements in the canonical construction.

Tree-Like Model Property

Theorem

For every $\varphi \in \mathcal{L}_{\text{WA-ATL}}$:

φ is satisfiable iff φ is satisfiable on a pointed tree-like CGMRP.

- Tree-like models have a unique root, unique predecessors, and no cycles.
- This lets us localize welfare labels along generated computations.
- Stability of preferences keeps those labels coherent over time.

Embedding into Standard ATL

Extend the atomic vocabulary with:

$$\heartsuit_i \quad \text{and} \quad \spadesuit_i$$

read as “agent i is benefited” and “agent i is harmed”.

Translation:

$$tr(\langle\langle C:B,H \rangle\rangle X\varphi) = \langle\langle C \rangle\rangle X(\heartsuit B \wedge \spadesuit H \wedge tr(\varphi)),$$

where

$$\heartsuit B = \bigwedge_{i \in B} \heartsuit_i, \quad \spadesuit H = \bigwedge_{i \in H} \spadesuit_i.$$

Similar clauses handle G and U.

Embedding Theorem and Complexity

Polynomial embedding

Satisfiability for WA–ATL reduces polynomially to satisfiability for standard ATL, with additional constraints ensuring:

best labels exist, worst labels exist, and no state is both for the same agent.

Complexity

Language	Target	Satisfiability
WA–ATL	ATL	EXPTIME-complete
WA–CL	CL	PSPACE-complete

The bounds are tight because ATL and CL are special cases.

Take-Home Message

- Strategic agency can be welfare-affecting: agents can benefit or endanger others through their strategies.
- WA–ATL internalizes this idea in the object language:

$$\langle\langle C:B, H \rangle\rangle\gamma.$$

- Potential benefactor and danger become definable notions.
- The logic remains technically well-behaved:
 - complete axiomatization for the safety fragment;
 - polynomial embedding into standard ATL/CL with welfare-label atoms;
 - tight complexity bounds.

Next steps

Full axiomatization with until; probabilistic risk-based danger; rational harmful strategies and strong danger.

Thank you

Questions?